

Center correlation breaks iid intervals

01/05

The point estimate targets β_1 , but the uncertainty model changes.

Synthetic multi-center simulation; treatment effect $\beta_1=0.25$.

$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_j + \varepsilon_{ij}$$

$$u_j \sim N(0, \tau^2), \quad \varepsilon_{ij} \sim N(0, \sigma^2)$$

$$\rho = \frac{\tau^2}{\tau^2 + \sigma^2}$$

ρ is the intraclass correlation;
center j is the inference unit.

β_1 is the treatment effect.

u_j : center-level shift shared within
center j .

ε_{ij} : individual residual noise.

Point estimates can remain centered while iid intervals lose coverage.

Grid: G=8/20/50 centers, ICC $\rho=0/.1/.3/.5$, balanced and imbalanced cluster sizes.

The sandwich variance aggregates by center

The middle term is a center-level score covariance, not a row-level residual pool.

Both intervals estimate β_1 ; only the variance object changes.

$$\hat{V}_{\text{CR}} = (X^\top X)^{-1} \left(\underbrace{\sum_g X_g^\top \hat{u}_g \hat{u}_g^\top X_g}_{\downarrow} \right) (X^\top X)^{-1}$$

Cluster meat: score contributions are summed by center before forming covariance.

$$\hat{V}_{\text{iid}} = \hat{\sigma}^2 (X^\top X)^{-1}$$

The simulation uses cluster-robust z intervals; CR2 and wild bootstrap remain planned comparisons.

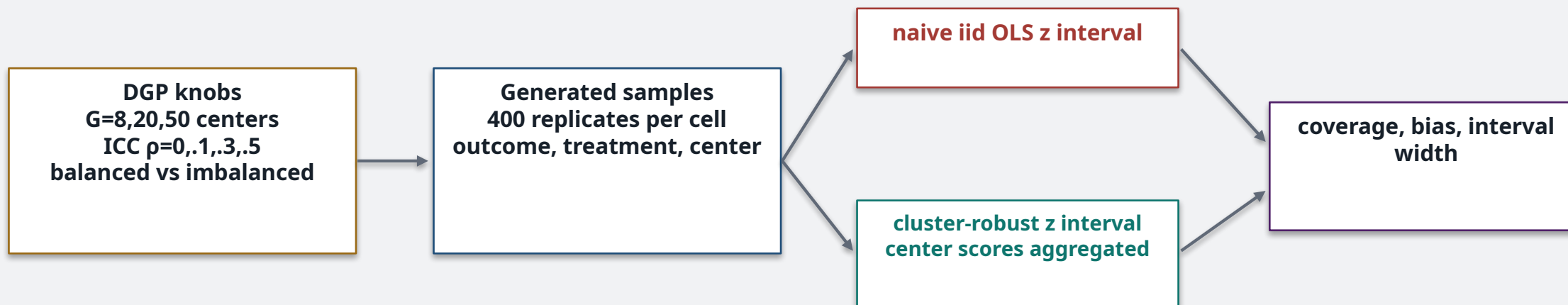
Naive variance treats rows as independent.

A stress grid separates interval behavior

03/05

The same generated samples feed two intervals and the same coverage endpoint.

Synthetic experiment: vary center count, ICC, and cluster-size balance.

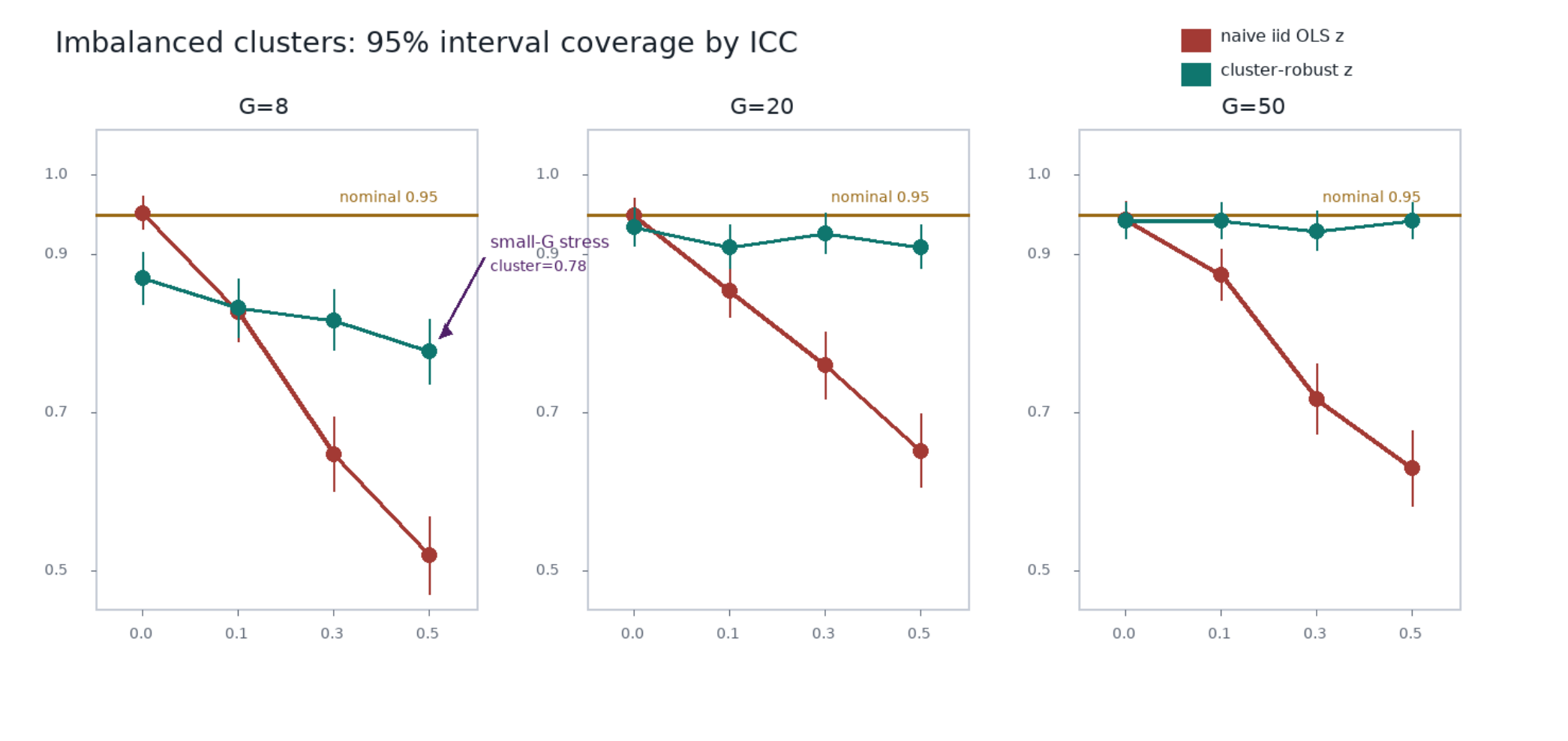


The design asks whether center aggregation repairs interval coverage across the stress grid.

Simulation study; endpoints are computed from the deterministic output for the same DGP grid.

Cluster adjustment recovers coverage except at small G

Coverage near 0.95 is the target; uncertainty is part of the figure.



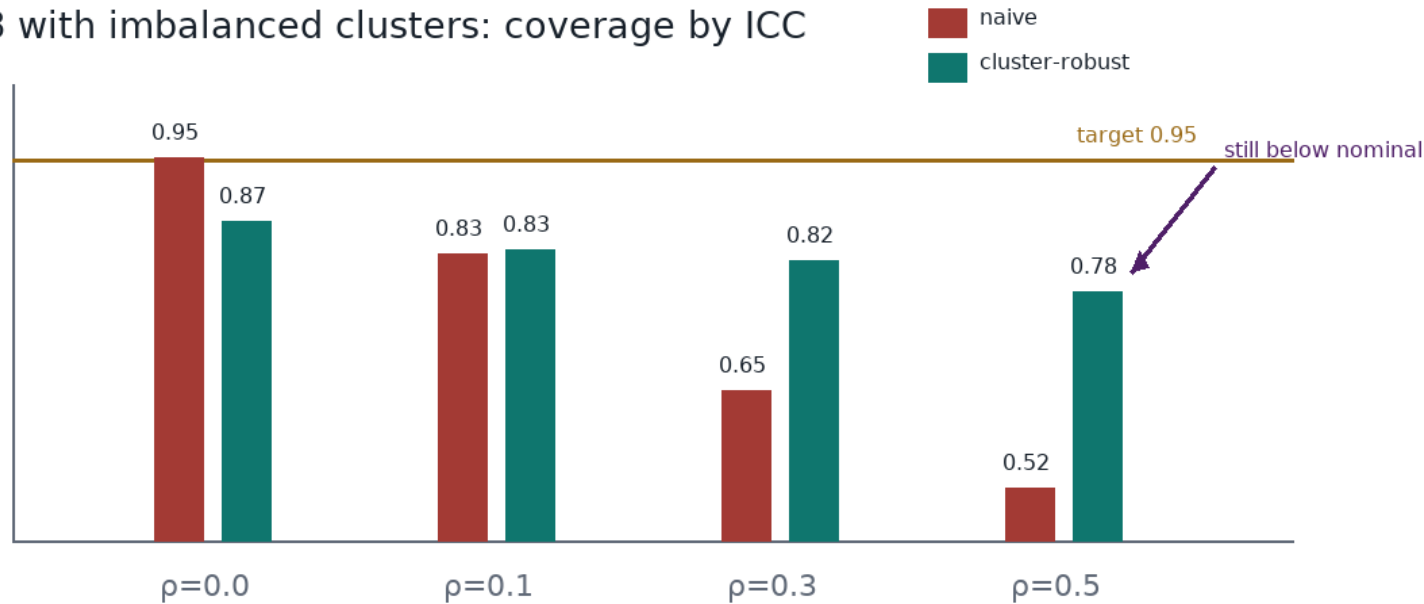
Synthetic imbalanced-cluster simulation; bands show Monte Carlo uncertainty.

The remaining failure is finite-center undercoverage

05/05

The next experiment tests small-sample corrections under the same grid.

G=8 with imbalanced clusters: coverage by ICC



At G=8, ICC $\rho=.5$, imbalanced clusters, cluster-robust coverage is 0.78.

Few centers make the sandwich variance noisy; imbalance increases center leverage.

Planned comparison: CR2 small-sample correction and wild cluster bootstrap on the same grid.

Completed evidence is limited to the two intervals shown; the correction methods are future tests.